

| Fast Motion | ★         | ★ ★ ★ ★   | ★ ★ ★ ★ ★ |
|-------------|-----------|-----------|-----------|
| Speed       | ★ ★ ★ ★ ★ | ★ ★ ★ ★ ★ | ★ ★ ★ ★ ★ |
| Simplicity  | ★ ★ ★ ★ ★ | ★ ★ ★ ★   | ★ ★ ★     |

## 5. Future Directions

- **Adaptive Kalman Filters:** Learn noise parameters online
- **Enhanced Appearance Models:** Multi-feature Mean Shift (color + texture)
- **Deep Learning Integration:** Use CNNs for features, classical methods for tracking
- **Non-linear Filters:** UKF and EKF for complex motion

## 🎯 FINAL VERDICT

### Mean Shift + Kalman Filter = Perfect Partnership

Use KF for motion prediction and occlusion handling

Use MS for precise appearance-based localization

Together they create robust, real-time tracking systems that work in challenging real-world scenarios!



## Research Paper Details